

Decision Making Under Uncertainty: 95-760

Recitation Session Week 6

PRACTICE PROBLEM FOR SUBMISSION

Instructions on how to submit: The problems can be handwritten or typed and should be uploaded to Gradescope by Saturday, October 3rd at 11:59 pm. The submission problem will require you to modify the spreadsheet "Recitation W6.xlsx" available on Canvas. You should copy a screenshot of each page of the Excel file showing all the parts of your problem. You don't need to submit the Excel document you use to solve the problems.

FISHERY OPERATIONS IMPLEMENTATION

In class we covered the fishery operations problem. There is a small fishery with one fish boat. Every day, operating costs amount to \$10,000, and 3,500 fish get caught. Fish are sold back to the port, but prices and demand are uncertain and variable. The distribution of the demand and the price are shown on the table below.

Price		Demand	Probability
Distribution	Normal	0	0.02
Mean	\$3.65	1,000	0.03
St. Dev	\$0.20	2,000	0.05
		3,000	0.08
		4,000	0.33
		5,000	0.29
		6,000	0.20

- On the first tab of the Excel file, "Sampling" generate 15 observations from each of the distributions listed in the table. The parameters for each distribution are shown in parenthesis. You can use Excel formulas or the "Random number generation" tool to complete this item.
- In the second tab, "Fishery Operations" simulate the profit of the fishery for 70 days. Calculate the mean, standard deviation, and 95% confidence interval.
- On the third tab, "N analysis", the objective is to perform an analysis to understand how the average profit changes as we increase the number of samples n . Simulate 500 days. For each day calculate the average, standard deviation, and 95% confidence interval of the profit up to the day you are analyzing. That is, if you are on Day 30, your average, standard deviation, and confidence interval should account for the profit from day 1 to 30. After filling out the table, you should see the graph of the mean and confidence interval on the right.
- On the fourth tab, "Analysis on Q", use the "Data Table" what if analysis to calculate the profit for different values of Q . After filling out the table, you should see the graph of the profit as a function of Q , to the right.

ADDITIONAL PRACTICE PROBLEMS - NO SUBMISSION REQUIRED

PROBLEM 1 - PIEDMONT AIRLINE

You work for Piedmont airline and want to analyze the profits of the company. The company sells tickets for a 19-seat flight. The demand is uncertain and follows a discrete distribution displayed in the table below. Each ticket is sold for \$150. A customer who buys a ticket has a no-show probability of 0.1. If more than 19 customers show up, the company pays \$500 per excess customer.

14	15	16	17	18	19	20	21	22	23	24	25
0.03	0.05	0.07	0.09	0.11	0.15	0.18	0.14	0.08	0.05	0.03	0.02

- If the company decides to sell 22 seats for the flight, what is the expected profit? Create a simulation with 100 samples. Calculate the mean, standard deviation, and 95% confidence interval of the profit.
- Is 100 samples a good number of samples for this simulation? What value of n would you recommend?
- Use the number of samples in (b) to do an analysis on the number of seats that the airline should offer to customers. How many seats would you recommend to the company to offer? Briefly justify your answer.

PROBLEM 2 - FOOD BANK

You are working for a regional food bank that provides food to community organizations. The food bank needs to decide how much food to purchase before the beginning of the month to ensure that it can supply the organizations. However, the amount of food donated during the month is uncertain. The food bank wants to make purchasing decisions that minimize its expected total cost while ensuring that food can be distributed to the community organizations once the donations are received. The food bank distributes three types of food: Canned vegetables, rice, and cereal.

Before the beginning of the month, the food bank must decide how many cases of each type of food to purchase. During the month, the food bank receives donations. The amount of donated food depends on the donation scenario. The food bank has identified three possible scenarios, each with a probability of occurring. The table below shows the amount of food (in cases) that the food bank expects to receive under each scenario.

Scenario	Probability	Canned Vegetables	Rice	Cereal
1	0.40	100	150	80
2	0.35	200	100	150
3	0.25	50	80	50

The food bank serves four community organizations. Each organization has a different demand for the three types of food. Demand depends on the number of families that each organization serves and may vary depending on the scenario. The table below shows the demand (in cases) for each food type under each scenario.

Scenario	Organization	Canned Vegetables	Rice	Cereal
1	Organization 1	120	100	80
	Organization 2	100	120	100
	Organization 3	80	100	70
	Organization 4	100	80	90
2	Organization 1	140	120	100
	Organization 2	120	140	120
	Organization 3	100	120	90
	Organization 4	120	100	110
3	Organization 1	100	90	70
	Organization 2	80	100	80
	Organization 3	70	80	60
	Organization 4	90	70	80

The food bank has a limited purchasing budget of \$12,000 and warehouse space for at most 700 cases of food for the purchase food. Assume that each case occupies one unit of warehouse space. The purchasing cost for each type of food is shown in the table below.

Food Type	Purchasing Cost per Case
Canned Vegetables	\$12
Rice	\$10
Cereal	\$15

At the end of the month, when the organization has received all the donations, the food bank decides how much food to distribute to each community organization. If the available food is insufficient to meet demand, the food bank can purchase additional cases from an external supplier at a higher cost. The emergency purchasing cost is \$18 per case of canned vegetables, \$16 per case of rice, and \$22 per case of cereal. This additional purchase is taken from the organization's emergency funds, not from the initial budget of \$12,000.

Assume that the food bank must satisfy the demand of each organization in each scenario. The food bank wants to minimize its expected total cost, including initial purchasing costs and emergency purchasing costs.

- Formulate the problem as a two-stage stochastic optimization model to help the food bank determine how much food to purchase while minimizing expected total cost. Clearly define the scenarios, decision variables (first-stage and second-stage), constraints, and objective function. Explain how uncertainty is represented in your model.
- Solve the problem using Excel Solver.
- Present your results to the food bank.